Sapphire Language Specification and Automation Manual

Current syntax and automation behavior as implemented in the interpreter.

This manual describes the currently tested implementation in this repository and intentionally avoids claiming production-scale or distributed infrastructure.

It is a local runtime reference for learning, experimentation, scripting, and bounded automation work, not a verified multi-node training or cloud deployment platform.

Core semantics

Sapphire is a dynamically typed, tree-walk interpreted language with expressions, statements, functions, value access, and structured results. The implementation emphasizes local execution and straightforward scripting ergonomics.

Automation modules

Automation support includes OS interaction, file and process operations, scheduling, notifications, HTTP helpers, and command control. Access may require optional dependencies or system permissions.

AI and ML

AI prompt, classification, and extraction helpers and local ML primitives are exposed when the corresponding packages and hardware are present. Their availability is environment-dependent and should not be assumed.

Portability

The project packages a Windows distribution and a Python-based source runtime. The source environment can differ from the bundled Windows installer, especially when optional dependencies are not installed.

Current status summary

Sapphire version 1.0.5 is a Python-based local runtime and language with a working interpreter, bundled tools, and optional integrations. It is appropriate for learning, local automation, and experimental AI workflows. It is not presented as a distributed training stack, a production security-certified platform, or a globally managed service.

Results vary by machine, Python version, OS configuration, library availability, and workload. The documentation here reflects what is actually exercised in this repository and what remains a roadmap item.

Version 1.0.5. Reproduce local measurements with benchmarks/benchmark_runtime.py. Values depend on the machine and environment.